

### CASE 15.3: THE IMAGINATION TOY CORPORATION

Amy White is the director of marketing for the Imagination Toy Corporation (ITC). She just received a phone call from her boss indicating that the company's board of directors gave final approval for the production and marketing of the Mighty Morphin' Monkeys—a new product line of action play toys for ITC. Amy worked hard in developing the concept for this product line and is thrilled that her ideas will become reality. But this news also means that she must get busy developing the marketing and sales force training materials needed to launch this new product line successfully. Amy's boss wants to know how soon she can have the sales staff trained and equipped to handle the new line.

The development of marketing materials and training of the sales staff for the new product line constitute a project. Amy identifies 10 specific project tasks that need to be accomplished before she can roll out the marketing program for this product line. First, Amy needs to collect information about the details of the decisions made by the board of directors. She can start this task (task A) immediately, and she estimates that it will take 5 days to determine exactly which items and accessories will be included in the first offering of the product line. After she completes this task, she will request prototypes of all items from the engineering department (task B), which she expects will take 10 working days. While waiting for the prototypes, she can begin laying out the marketing program (task C). She expects this activity to take 8 days. After the prototypes (task B) are available, Amy estimates that it will take 7 days to prepare instructions (task D) on the operation and use of the items in the product line and 9 days to design its packaging (task E). When the marketing program (task C) is finished, it must be approved (task F) by the president of the company. Amy expects this approval to take 3 days.

Amy plans to hold a 2-day training course (task G) for the sales force once the operating instructions (task D) and the packaging design (task E) are completed. When the operating instructions (task D) are finished and the marketing plan is approved (task F), Amy will develop an information guide (task H) that the sales force can distribute to retailers. Amy expects to take 8 days to complete the information guide. Also, as part of the marketing plan, Amy wants to hire a number of actors to portray Mighty Morphin' Monkeys (task I) at various promotional events around the country. Hiring and training these actors are expected to take 8 days and can be done only after the marketing plan is approved (task F). Finally, after the marketing plan (task F) has been approved and the packaging for the product has been designed (task E), special point-of-sale display racks must be manufactured (task J). Amy expects this activity to take 12 working days.

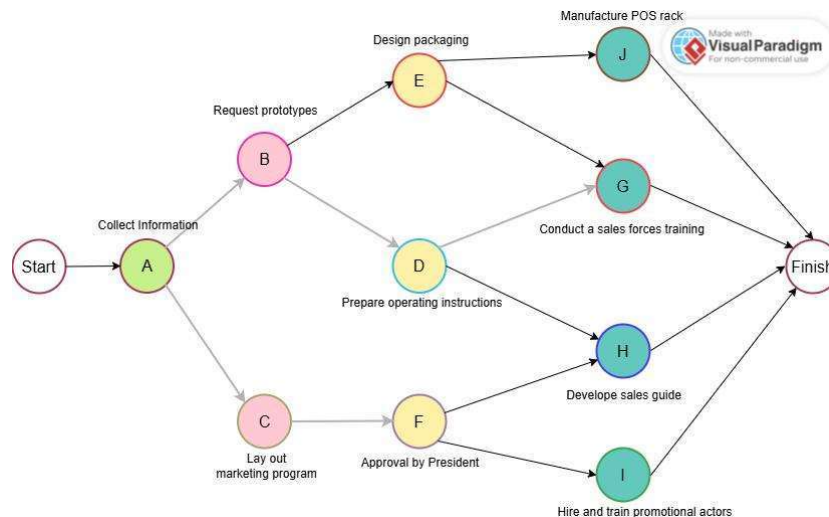
#### a, Develop an AON network for this problem

The questions require us to find the optimal path for the project, so here are the table for the activities list for this project:

| Task | Description                               | Time | Predecessor |
|------|-------------------------------------------|------|-------------|
| A    | Collect information about board decisions | 5    | — (Start)   |
| B    | Request prototypes from engineering       | 10   | A           |

|          |                                             |    |      |
|----------|---------------------------------------------|----|------|
| <b>C</b> | Lay out the marketing program               | 8  | A    |
| <b>D</b> | Prepare operating instructions              | 7  | B    |
| <b>E</b> | Design packaging                            | 9  | B    |
| <b>F</b> | Marketing program approval by the president | 3  | C    |
| <b>G</b> | Conduct a sales force training course       | 2  | D, E |
| <b>H</b> | Develop a sales information guide           | 8  | D, F |
| <b>I</b> | Hire and train promotional actors           | 8  | F    |
| <b>J</b> | Manufacture point-of-sale display racks     | 12 | E, F |

Based on the activities table, this is the AON network for this project.



**b. Create a spreadsheet to calculate and summarize the earliest and latest start and finish times, the slack for each activity, and the critical activities.**

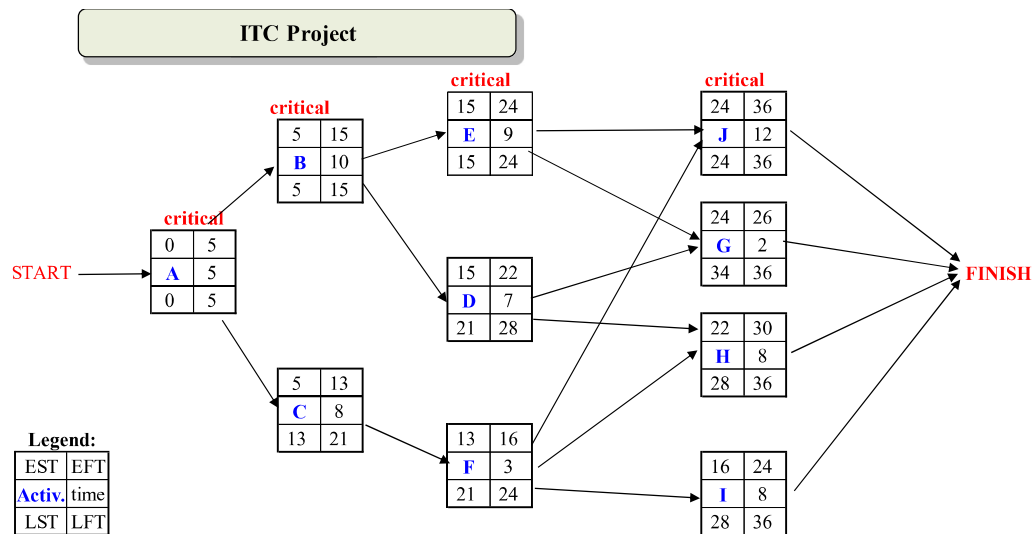
So now, we will use the spreadsheet to find the earliest and latest start and finish times, the slack for each activity, and the critical activities. Based on the time required for each task, we have the formula:

$$EFT = EST + t$$

$$LST = LFT - t$$

| Activity | Description                                 | Predecessor | Time      | EST | EFT | LST | LFT | Slack     |
|----------|---------------------------------------------|-------------|-----------|-----|-----|-----|-----|-----------|
| <b>A</b> | Collect information about board decisions   |             | <b>5</b>  | 0   | 5   | 0   | 5   | <b>0</b>  |
| <b>B</b> | Request prototypes from engineering         | A           | <b>10</b> | 5   | 15  | 5   | 15  | <b>0</b>  |
| <b>C</b> | Lay out the marketing program               | A           | <b>8</b>  | 5   | 13  | 13  | 21  | <b>8</b>  |
| <b>D</b> | Prepare operating instructions              | B           | <b>7</b>  | 15  | 22  | 21  | 28  | <b>6</b>  |
| <b>E</b> | Design packaging                            | B           | <b>9</b>  | 15  | 24  | 15  | 24  | <b>0</b>  |
| <b>F</b> | Marketing program approval by the president | C           | <b>3</b>  | 13  | 16  | 21  | 24  | <b>8</b>  |
| <b>G</b> | Conduct a sales force training course       | D, E        | <b>2</b>  | 24  | 26  | 34  | 36  | <b>10</b> |
| <b>H</b> | Develop a sales information guide           | D, F        | <b>8</b>  | 22  | 30  | 28  | 36  | <b>6</b>  |
| <b>I</b> | Hire and train promotional actors           | F           | <b>8</b>  | 16  | 24  | 28  | 36  | <b>12</b> |
| <b>J</b> | Manufacture point-of-sale display racks     | E, F        | <b>12</b> | 24  | 36  | 24  | 36  | <b>0</b>  |

### MODEL NETWORK:



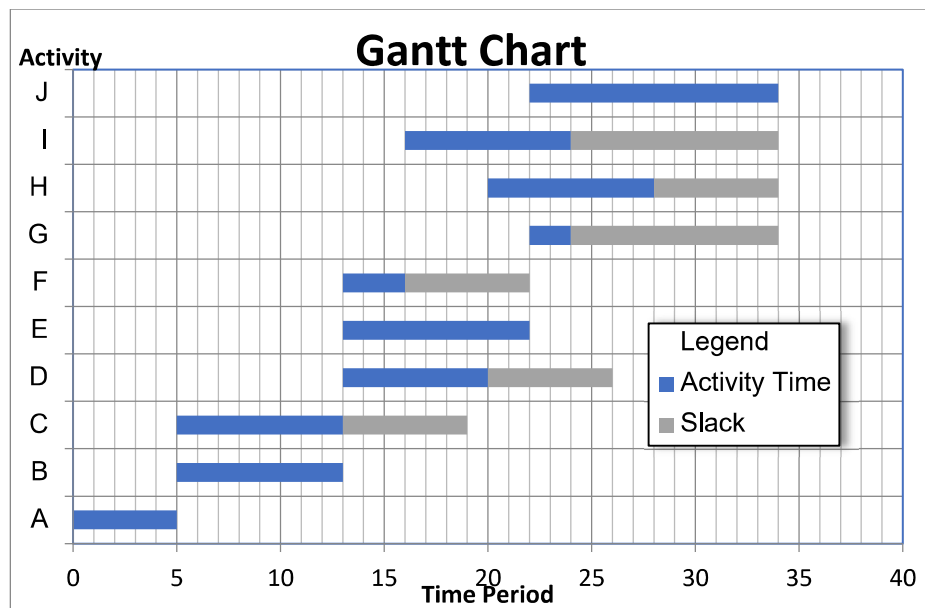
### c. Identify the critical path.

Based on the table, we can immediately see that the critical path is:

**A → B → E → J**

These tasks are critical as if they are not finished on time, the whole project will be delayed.

Here is the Gantt Chart that clearly shows the time for each task and its slack.



d. If Amy starts working immediately, how long will it take her to complete this project?

If Amy starts working immediately, it will take her **36 days** to complete.

e. Suppose that the engineering department can create the prototypes in only 8 days if the engineers work overtime on this activity. Would it help reduce the length of time required to complete this project?

Suppose that the engineering department can create the prototypes in only 8 days if the engineers work overtime on this activity. From the previous findings, we know that B is the critical activity, so that reducing the days needed to do it will affect the overall progress for this project. So, we had changed the activity B time to 8 days to see what happened.

The new spreadsheet after changing the prototypes from engineering:

| Activity | Description                                 | Predecessor | Time      | EST | EFT | LST | LFT | Slack     |
|----------|---------------------------------------------|-------------|-----------|-----|-----|-----|-----|-----------|
| <b>A</b> | Collect information about board decisions   |             | <b>5</b>  | 0   | 5   | 0   | 5   | <b>0</b>  |
| <b>B</b> | Request prototypes from engineering         | A           | <b>8</b>  | 5   | 13  | 5   | 13  | <b>0</b>  |
| <b>C</b> | Lay out the marketing program               | A           | <b>8</b>  | 5   | 13  | 11  | 19  | <b>6</b>  |
| <b>D</b> | Prepare operating instructions              | B           | <b>7</b>  | 13  | 20  | 19  | 26  | <b>6</b>  |
| <b>E</b> | Design packaging                            | B           | <b>9</b>  | 13  | 22  | 13  | 22  | <b>0</b>  |
| <b>F</b> | Marketing program approval by the president | C           | <b>3</b>  | 13  | 16  | 19  | 22  | <b>6</b>  |
| <b>G</b> | Conduct a sales force training course       | D, E        | <b>2</b>  | 22  | 24  | 32  | 34  | <b>10</b> |
| <b>H</b> | Develop a sales information guide           | D, F        | <b>8</b>  | 20  | 28  | 26  | 34  | <b>6</b>  |
| <b>I</b> | Hire and train promotional actors           | F           | <b>8</b>  | 16  | 24  | 26  | 34  | <b>10</b> |
| <b>J</b> | Manufacture point-of-sale display racks     | E, F        | <b>12</b> | 22  | 34  | 22  | 34  | <b>0</b>  |

So instead of 36 days, the project now only takes **34 days** to complete.